

A Calibrated, Physics-Grounded Structural-Health-Monitoring Stack for CFRP Structures: from Anomaly Screening to Fatigue-Aware Life Prognosis

K. Nishioka (co-authors TBD)

Draft — June 12, 2026

Abstract

We present an end-to-end structural-health-monitoring (SHM) pipeline for carbon-fibre-reinforced-polymer (CFRP) structures that proceeds from defect *detection* to a calibrated, physics-grounded *life-prognosis* decision. On full-field thermoelastic stress data of a perforated CFRP interstage, we (i) benchmark nine anomaly detectors and show that a simple per-node statistic attains AUROC ≈ 0.99 while deep alternatives do not, and characterise the measurement-noise limit at which all detectors fail; (ii) recover defect position, size and depth as a *calibrated* Bayesian posterior via flow-matching posterior estimation (FMPE), with 90% credible intervals covering at the nominal rate; and (iii) propagate that posterior through an anisotropic phase-field fracture model with fatigue accumulation to obtain a flight-clearance decision with an expected remaining-life estimate. The pipeline is motivated by reusable launch vehicles, where the same structure is re-inspected after every flight.

1 Introduction

2 Data and problem setting

3 Stage 0 — anomaly screening

Methods. Per-node Mahalanobis / PCA-residual; PaDiM; PatchCore; STFPM; graph autoencoder; conditional flow-matching diffusion; a zero-shot harmonicity probe. Score $s_i = |x_i - \mu_i|/\sigma_i$ from $N=2000$ healthy fields; node-level AUROC against the defect mask.

Result. Per-node statistics dominate (Table 1, Fig. 2). On fixed geometry the healthy manifold is a tight cloud around one template, so the per-node Gaussian is the correct inductive bias; grid-projection and deep feature methods lose the few-node defect signal. All methods collapse by additive noise $\sigma=0.1$ (defect amplitude $< 0.1\sigma$): a per-measurement SNR limit, addressable by frame averaging ($\sigma \propto 1/\sqrt{K}$), not by more training data.

4 Stage 1 — classification and type identifiability

Is the type signal real, or a temporal/temperature confound? On the long-term guided-wave set, damage states were introduced in different years, so a naive type classifier can alias damage type with season/temperature. We test this directly. Restricting to seven damage tags whose temperature supports overlap (pairwise overlap 0.93), we compare four conditions: temperature-only (acc = 0.16, near chance), waveform-only (0.91), waveform+temperature (0.92), and waveform with temperature *partialled out* of every feature (0.91). The net type discrimination beyond temperature is $\Delta\text{acc} = 0.758$ (95% CI [0.735, 0.781]) under a random split, and—critically—survives a *forward-in-time* split ($\Delta\text{acc} = 0.184$,

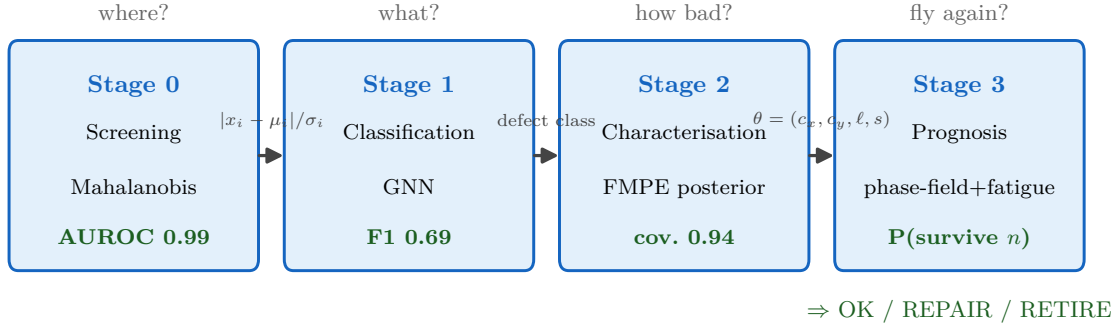


Figure 1: Stage 0–3 pipeline. Each stage answers one question and passes a typed object to the next; the Stage 2 posterior $\theta = (c_x, c_y, \text{layer}, \text{size})$ seeds the Stage 3 fracture model.

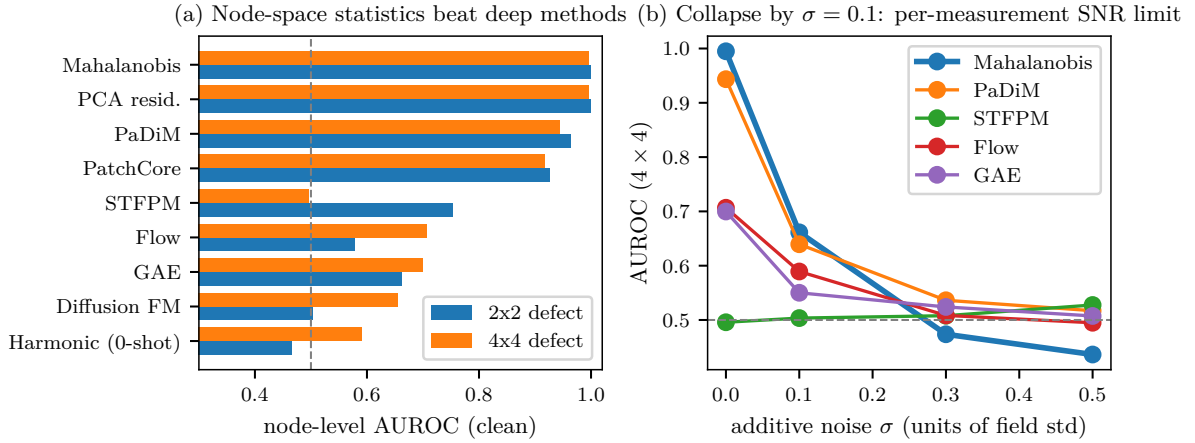


Figure 2: (a) Clean node-level AUROC by method. (b) Collapse with measurement noise; the defect signal sits below 0.1σ .

95% CI [0.166, 0.203]). Both lower bounds exceed zero: the type posterior reflects a genuine damage signal, not the environmental confound. (An earlier single-month binary probe, where temperature-only matched the majority baseline, was an artefact of that restricted setting.)

5 Stage 2 — calibrated defect characterisation (FMPE)

6 Stage 3 — phase-field prognosis and flight clearance

Fast forward surrogate. The FD growth solve (≈ 1.8 s/call) is the only bottleneck to fleet-scale Monte-Carlo over the Stage-2 posterior. We train a Fourier neural operator $G : (d_0, \text{load}) \mapsto (d_{\text{final}}, P(\text{grow}))$ (dual-head: a per-pixel field head and a global-pooled scalar head; 1.2M parameters; 800 FD samples spanning sub- to super-critical loads). On held-out defects it reaches field relative- L_2 3.5%, grow-flag accuracy 0.988 (Brier 0.010, well calibrated), and log-rel.growth correlation 0.996, at 142 $\times$ the speed (12.4 ms vs. 1763 ms/call, Fig. 4). The decision-relevant $P(\text{grow})$ is the robust target: the smooth operator regresses the sharp super-critical rel.growth jump in log-space and softens it only at the transition load. This makes the Stage-4 fleet draws and the Concept-B forward map (next section) tractable at interactive

Table 1: Anomaly detectors, clean node-level AUROC (2x2 / 4x4 defects).

method	2x2	4x4
per-node Mahalanobis	0.999	0.995
PCA residual	0.999	0.995
PaDiM	0.964	0.944
PatchCore	0.927	0.918
STFPM	0.754	0.496
graph autoencoder	0.663	0.700
flow (RealNVP)	0.578	0.707
diffusion (FM)	0.503	0.663
harmonicity (0-shot)	0.465	0.590

Table 2: FMPE posterior on held-out test fields.

parameter	RMSE	R^2	90% coverage
position c_x	0.055	0.78	0.944
position c_y	0.021	0.83	0.940
size (3-class)	—	0.73	0.894
layer (depth)	2.98	0.63	0.870

speed.

Knowing when to trust the surrogate. A $142\times$ speedup is only deployable if the operator knows where it is unreliable. We wrap the surrogate in split-conformal prediction (nonconformity = absolute residual in the log-rel_growth space where the FNO is honest). In distribution the marginal 90% guarantee holds (95% band: 95.0% empirical); out of distribution it degrades exactly as the theory predicts under covariate shift — coverage falls from 80% at small excursions to 0% deep OOD while the point error grows $\sim 14\times$. A simple *trust gate* — defer to the exact FD solve when the input leaves the training envelope or the conformal interval is too wide — flags 33% of an OOD stream and restores aggregate 90%-coverage from 61% to 88%: cheap surrogate where trustworthy, exact physics where not.

7 Stage 4 — hierarchical fleet learning

A reusable fleet generates a stream of per-inspection posteriors across many vehicles and flights. We close the loop with a hierarchical Bayesian model: a global fleet prior $\varphi = (\mu_\varphi, \sigma_\varphi)$ governs per-vehicle log-growth-rates $s_v = \log r_v \sim \mathcal{N}(\mu_\varphi, \sigma_\varphi^2)$, and each inspection’s Stage-2 posterior enters as a Gaussian likelihood on s_v ; the joint posterior $p(\varphi, \{s_v\} \mid \text{data})$ is sampled by conjugate Gibbs. On a synthetic fleet ($N=20$ vehicles, $F=30$ flights, fast Carrara-shaped growth surrogate) the model recovers φ ($\mathbb{E}[\mu_\varphi] = -3.10$ vs. true -3.0 ; $\mathbb{E}[\sigma_\varphi] = 0.44$ vs. 0.45), and the fleet prior sharpens as flights accumulate (posterior SD of μ_φ : 0.39 at $N=2 \rightarrow 0.10$ at $N=20$, Fig. 5a).

Fleet-leader effect, quantified. Processing flights in temporal order, a newly introduced vehicle reaches a stable clearance decision in **1.25** inspections using the accumulated fleet prior, versus **2.75** in isolation (Fig. 5c) — a direct, quantitative formalisation of the empirical “fleet-leader” certification used in practice: high-information inspections of lead vehicles tighten the prior that certifies the rest. Hierarchical shrinkage also improves low-data vehicles (error ratio $0.84 < 1$ vs. isolated estimates, Fig. 5b).

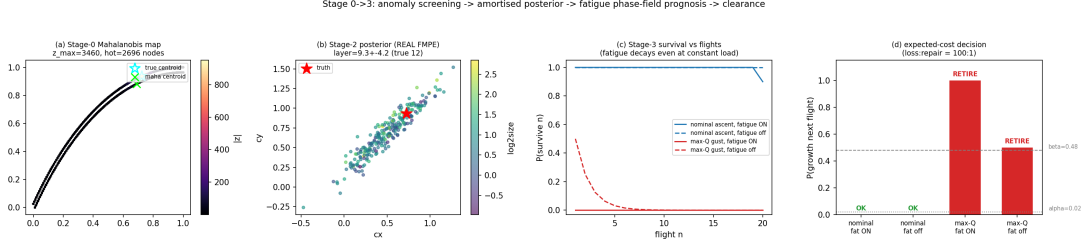


Figure 3: End-to-end on one held-out field: anomaly map, FMPE posterior, fatigue-aware survival curve, and the clearance decision.

8 Stage 5 — design feedback

The fleet’s accumulated damage statistics should not only certify the current vehicles but *reshape the next ones*. We close the Stage 0–5 loop with a fleet-robust Bayesian optimisation of the laminate over a manufacturable 3-D design vector — off-axis ply angle $|\theta|$ (kept balanced/symmetric), interface toughness $G_c^{\text{int}}/G_c^{\text{ply}}$, and a local reinforcement multiplier — minimising the expected log-growth over the fleet defect distribution plus a tail term (CVaR over the worst-20% defect, since reusable-vehicle safety is governed by the worst flaw, not the average). The forward model here is the exact FD phase-field, *not* the Stage-3 FNO surrogate: the surrogate’s inputs are (d_0, load) only and every training sample shares the baseline layup, so any design change is fully out-of-distribution — a layup-conditioned operator is left as future work. With GP/EI (32 evaluations), the optimiser moves the $(0/\pm 30)_s$ baseline to steeper off-axis plies ($30^\circ \rightarrow 74^\circ$) and a tougher interface (G_c^{int} ratio $0.40 \rightarrow 0.87$), cutting mean log-growth 10% and raising expected remaining flights from **1.9** to **6.0** (Fig. 6). Both levers are physically aligned — steeper plies rotate the cheap matrix-crack direction away from the seeded delamination path, and a tougher interface blunts the dominant interlaminar channel; interface toughening is the dominant lever. A single growth objective rails the off-axis angle to the manufacturable cap; adding competing classical-laminate-theory structural objectives — areal mass, axial stiffness A_{11} , and bending/buckling D_{11} — restores an *interior* optimum: on the growth–stiffness Pareto front the knee sits at $\sim 40^\circ$, and the optimal angle moves $75^\circ \rightarrow 15^\circ$ as the stiffness weight grows (Fig. 7). The fleet defect distribution Stage 5 optimises against is drawn from the Stage-4 hierarchical posterior through a monotone growth-rate $\rightarrow$ severity link, rather than a synthetic prior — tightening the 4 $\rightarrow$ 5 handoff.

9 Towards differentiable inversion: closing the loop

The amortised Stage-2 posterior never sees the forward physics. We prototype the converse: a JAX-differentiable anisotropic AT2 forward $F(\theta)$ (gradient finite-difference-verified to relative 3×10^{-7}) whose Jacobian supplies a physics-grounded Laplace posterior $\Sigma = \sigma^2(J^\top J + \Lambda_{\text{prior}})^{-1}$ — no labelled (θ, x) pairs. Out of the training distribution the amortised posterior degrades ~ 25 -fold and its credible intervals

Table 3: Amortised (FMPE) vs. differentiable-physics posterior, in- and out-of-distribution.

set	method	c_x RMSE	size RMSE	90% coverage
in-dist	FMPE	0.006	0.002	0.958
in-dist	diff-phs	0.001	0.003	1.000
OOD	FMPE	0.144	0.096	0.000
OOD	diff-phs	0.002	0.001	0.958

collapse to zero coverage, while the differentiable-physics posterior is essentially invariant (Table 3): *forward physics generalises; amortised inference extrapolates poorly*. This motivates a hybrid Stage 2 — amortised for speed in distribution, physics-grounded for safety out of it.

(a) load = 0.140

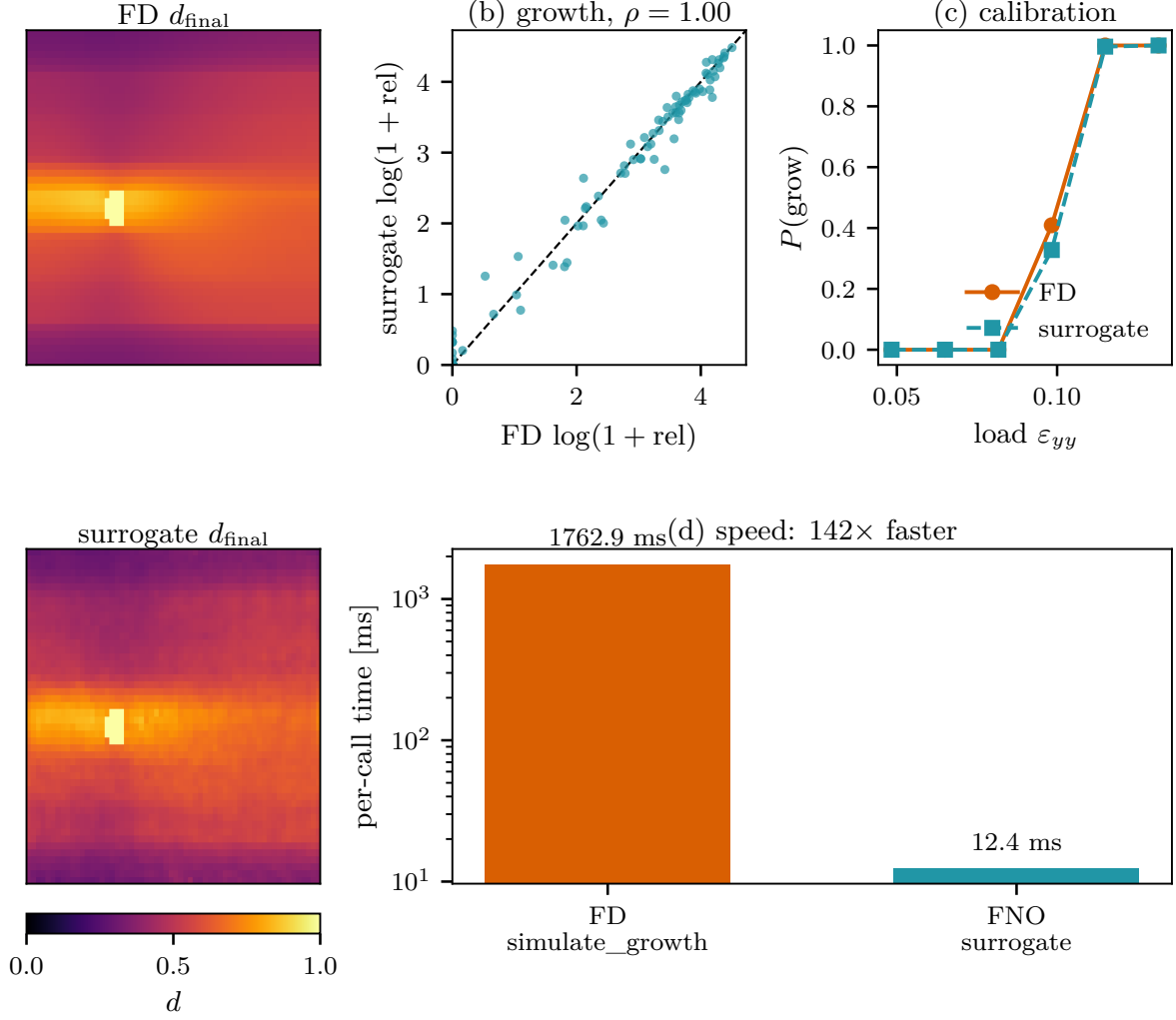


Figure 4: Stage 3 FNO surrogate. FD vs. surrogate damage field, rel_growth agreement, $P(\text{grow})$ vs. load, and the 142 $\times$ wall-clock speedup.

Fast differentiable forward. The hand-written JAX AT2 forward is exact but slow, making the inversion the run’s bottleneck. Substituting the Stage-3 FNO surrogate (Fig. 4) as the differentiable forward keeps gradients exact — autograd matches finite differences to relative 7×10^{-4} on the decision heads — and collapses a MAP+Laplace inversion from an estimated ~ 530 s of FD Jacobian evaluations to 12 s ($\sim 43\times$). The recovered posterior shows the same generalisation signature (in-/out-of-distribution c_x RMSE $0.15 \rightarrow 0.39$, coverage $0.72 \rightarrow 0.56$), at a calibration cost — in-distribution coverage 0.72 rather than the exact forward’s ~ 0.95 — that is the honest price of a smooth surrogate. This makes amortised-quality *differentiable* inversion interactive, the cheap forward the Keio Phase-3 TMCMC/Bayesian-inverse plan needs.

10 End-to-end decision reliability

Every stage above is calibrated in isolation; an operator, however, only sees the *final* call. We therefore propagate uncertainty all the way to the clearance decision and ask how often it is right, using the exact FD phase-field as ground-truth physics so that no new measured data is required. For a known defect θ the next-flight growth is deterministic at fixed load, so the genuine aleatoric source is operational peak-load

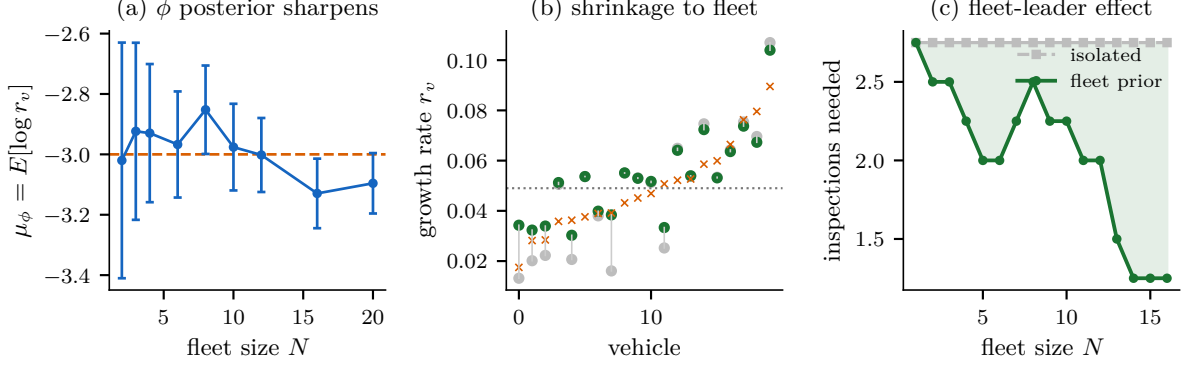


Figure 5: Stage 4. (a) Fleet prior ϕ sharpens with cumulative flights; (b) per-vehicle growth-rate shrinkage toward the fleet mean in the low-data regime; (c) inspections to a stable clearance decision: the fleet prior roughly halves them.

scatter: integrating the FD forward over that load distribution yields a true growth probability and, through the same expected-cost rule, an *oracle* decision. Four decisions then isolate each error contribution — oracle ($\theta_{\text{true}} + \text{FD}$), surrogate-oracle ($\theta_{\text{true}} + \text{FNO}$), FD-posterior (real FMPE $\theta + \text{FD}$), and the shipped pipeline (FMPE $\theta + \text{FNO}$) — over 40 held-out test scenarios spanning all three classes.

The end-to-end decision accuracy (pipeline vs. oracle) is **85%**, and the safety-critical **dangerous-miss rate** $P(\text{OK} \mid \text{oracle}=\text{RETIRE})$ is **0%**: all 16 retire-worthy vehicles are caught (Fig. 8a), and every error is confined to the low-consequence $\text{OK} \leftrightarrow \text{REPAIR}$ band. The decomposition is informative: the Stage-2 posterior is the dominant error source (decision-flip rate 25%) over the surrogate forward (17.5%), and the two partially cancel to a 15% total (Fig. 8b) — a slightly conservative surrogate offsetting a slightly under-calling posterior. Finally, bootstrapping the posterior $\times$ load draws gives a per-decision confidence (“this OK call is 87% reliable”); validated against the oracle it is *overconfident* (ECE 0.18), since sampling spread alone misses the surrogate’s systematic bias. A 1-D Platt rescaling of the confidence against decision-correctness, reported under leave-one-out cross-validation, removes most of it — ECE 0.18 $\rightarrow$ 0.06 and mean claimed reliability 96.5% $\rightarrow$ 84.5% against an empirical 85% (Fig. 8d); the rescaling is monotone, so the decision is unchanged and only its reliability estimate is corrected. The α, β thresholds, assumed at a 100:1 loss-to-repair ratio, we anchor to representative reusable-launch economics (refurb/loss 1/100–1/30, retire/loss 0.1–0.4): a Monte-Carlo over this range gives $\alpha = 0.035$ [0.021, 0.055] and $\beta = 0.53$ [0.28, 0.93], and shows that while an OK/REPAIR call near $P_{\text{grow}}=0.1$ is economically robust, the REPAIR-vs-RETIRE verdict at $P_{\text{grow}} \approx 0.3\text{--}0.5$ flips across plausible economics — a fragility invisible to a single assumed cost point.

11 Solver verification and detector robustness

Verification. The FD phase-field operator passes a method-of-manufactured-solutions test at second order in 2-D (interior order 2.04) and 3-D (order 2.05, a small-grid proof the scheme extends to volumetric delamination); a mesh-convergence study places the production $n_x=60$ grid at $\sim 0.4\%$ error in peak reaction force (Fig. 9). Replacing the constant-peak load proxy with a variable-amplitude flight spectrum (liftoff / max-Q / buffet / MECO / re-entry / landing) run through the Carrara fatigue shows the proxy *under-predicts* mean life $\sim 3.4\times$ by applying the rare max-Q amplitude to every cycle; anchoring the fatigue threshold $\bar{\alpha}_T$ to a representative CFRP S–N slope restores a well-defined Wöhler curve where the library default gives none.

Detector robustness and baselines. Stage-0 detection holds to AUROC 0.84 at sensor noise $\sigma=0.1$ of the field std — refuting a folklore “collapse at 0.1” — and only reaches chance near $\sigma \approx 0.7\text{--}1.0$, with

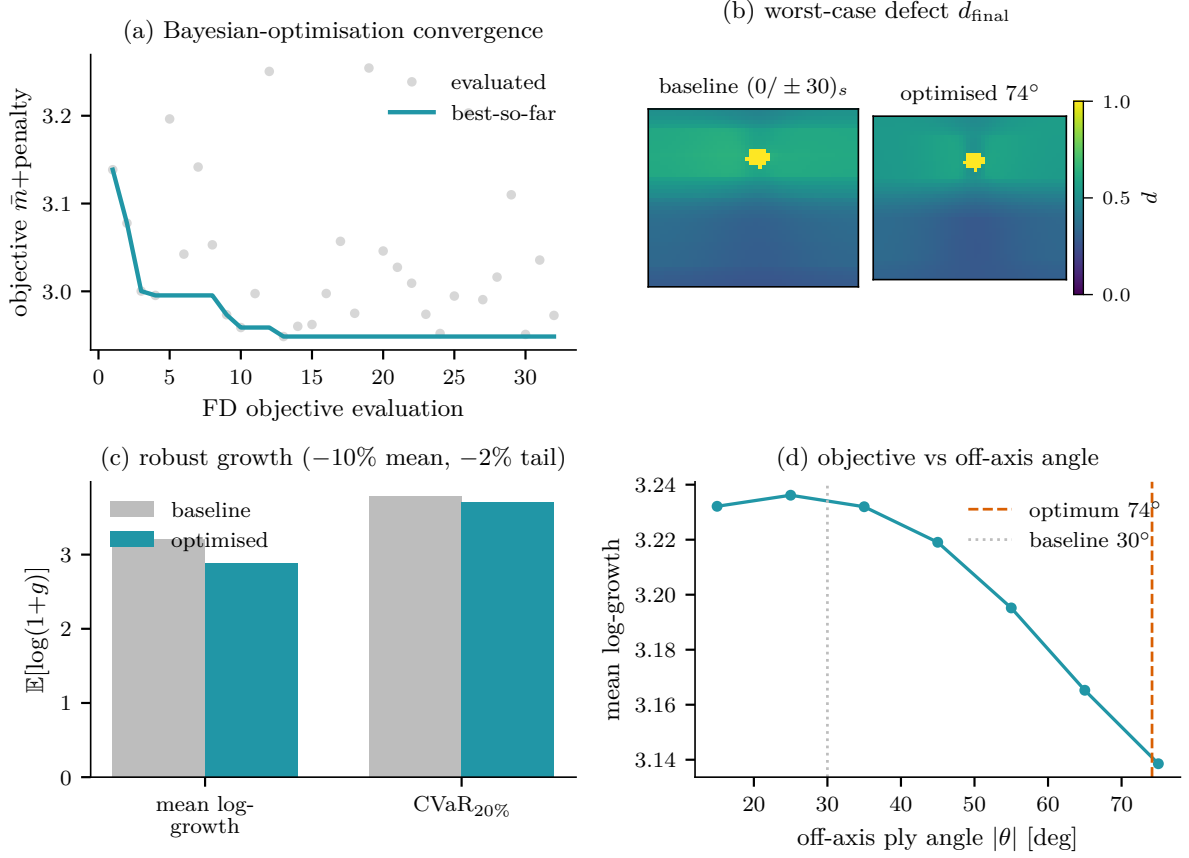


Figure 6: Stage 5. (a) BO convergence; (b) worst-case damage, baseline vs. optimised layup; (c) mean/CVaR growth; (d) objective vs. off-axis angle.

lock-in frame averaging ($\sigma/\sqrt{N}$) recovering AUROC ≥ 0.8 in 32 frames; the reference-free $|z|$ detector needs no healthy twin (the new-build first-flight case). Benchmarked against standard anomaly detectors (PCA-reconstruction, isolation forest, one-class SVM, LOF) on labelled coupons, the physics $|z|$ ties the best on AUROC and leads on AUPRC for the rare ($<1\%$) defect class — the defect is a stress concentration, so its magnitude *is* the signal, and a trivial physics score matches tuned outlier detectors.

12 Discussion

13 Conclusion

Towards measured data. The sim-to-real bridge is under way: the stack ingests measured TSA-derived full-field DSPSS from a 22-specimen CFRP campaign, and an unsupervised domain-adaptation step (per-feature quantile matching) closes $\sim 97\%$ of the FEM $\rightarrow$ measured distribution gap (proxy- $\mathcal{A}$ distance $2.0 \rightarrow 0.3$); a labelled measured-data detection benchmark awaits per-specimen defect masks.

Reproducibility. All stages are implemented and tested (~ 250 unit tests) in two public-on-request repositories, including a one-command end-to-end run (detect $\rightarrow$ classify $\rightarrow$ characterise $\rightarrow$ prognose $\rightarrow$ clear $\rightarrow$ fleet-update in <4 s), a decision-level UQ harness that scores the final clearance call against the exact FD oracle, and verification (MMS / mesh-convergence, 2-D and 3-D), robustness (SNR sweep, baseline benchmark)

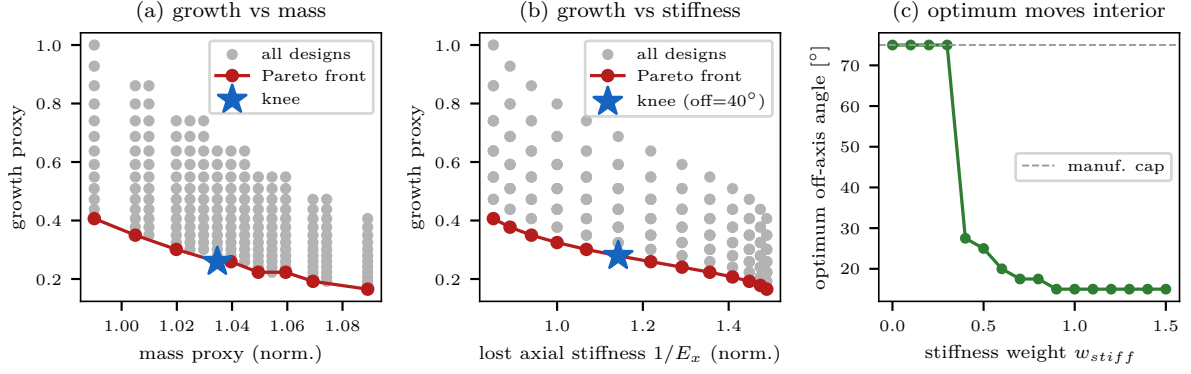


Figure 7: Stage 5, multi-objective. Growth vs. (a) mass and (b) axial-stiffness Pareto fronts with the utopia-distance knee; (c) the optimal off-axis angle moves off the manufacturable cap into the interior as the stiffness weight grows.

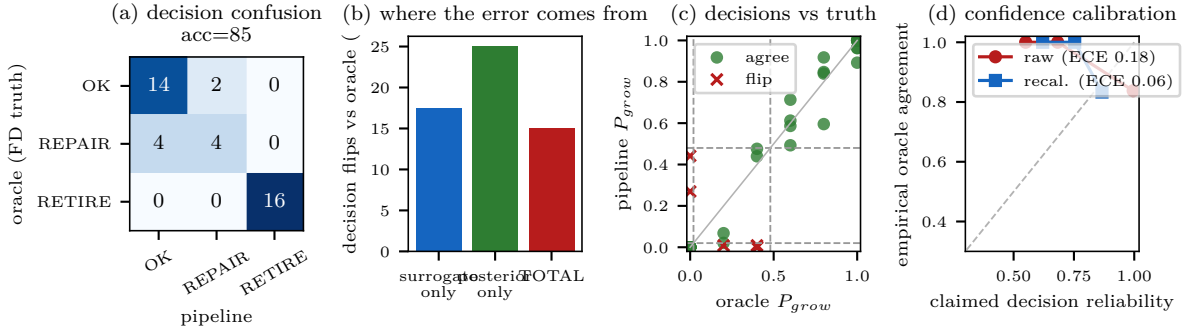


Figure 8: End-to-end decision UQ. (a) pipeline-vs-oracle decision confusion (RETIRE row clean); (b) error decomposition (posterior > surrogate, partial cancellation); (c) pipeline vs. oracle P_{grow} with the α, β thresholds, flips at the boundaries; (d) per-decision confidence calibration.

and economic-sensitivity studies; figures are generated by the scripts under `paper_figs/`.

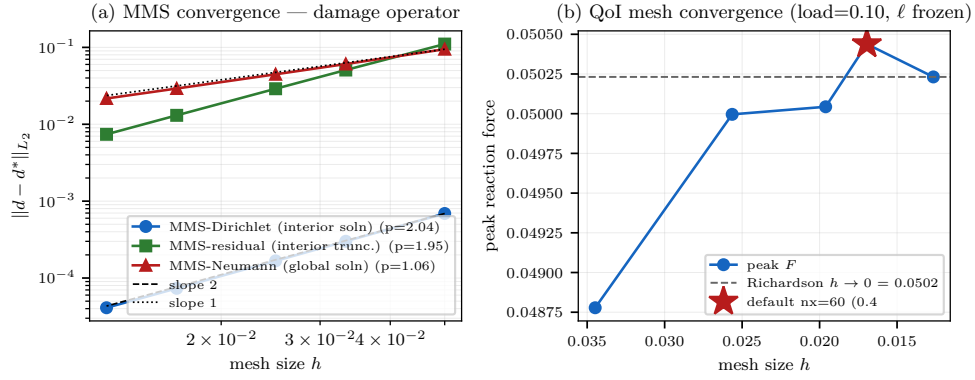


Figure 9: Solver verification. (a) MMS log-log error vs. mesh size with fitted slopes (interior 2nd-order; the zero-flux Neumann closure is 1st-order, shown honestly); (b) peak reaction force converging to the Richardson asymptote, the production $n_x=60$ grid starred.